

🐍 Python Developer Roadmap — Beginner

****1. Computer & Programming Basics****

→ Computer fundamentals → Programming concepts → Algorithms basics → VS Code → Terminal

****2. Python Fundamentals****

→ Syntax → Variables → Data types → Operators → Input/Output → Conditions → Loops → Strings

****3. Python Data Structures****

→ Lists → Tuples → Sets → Dictionaries

****4. Functions & Modules****

→ Functions → Parameters → Return → Scope → Lambda → Modules → Packages

****5. Files & Error Handling****

→ File handling → JSON → CSV → Exceptions → Debugging

****6. Object-Oriented Python****

→ Classes → Objects → Constructors → Methods → Inheritance → Encapsulation → Polymorphism → Abstraction

****7. Advanced Python****

→ Comprehensions → Iterators → Generators → Decorators → `\*args` → `\*\*kwargs` → Context managers

****8. Problem Solving & DSA****

→ Big-O → Arrays/Lists → Strings → Hashing → Stack → Queue → Searching → Sorting → Recursion

****9. Python Environment****

→ pip → Virtual environments → Packages → `requirements.txt`

****10. SQL & Databases****

→ Database fundamentals → SQL → CRUD → Joins → Subqueries → Aggregation → PostgreSQL → Database design

****11. Git & GitHub****

→ Git basics → Repository → Commit → Branch → Merge → Pull → Push → GitHub → README

****12. Web Fundamentals****

→ Internet → Client/Server → HTTP/HTTPS → Request/Response → JSON → REST → APIs
→ HTML/CSS basics

****13. Backend Development****

→ FastAPI → Routes → HTTP methods → Pydantic → Validation → REST APIs →
Swagger/OpenAPI

****14. Database Integration****

→ SQLAlchemy → ORM → Models → CRUD → Relationships → Migrations

****15. Authentication & Security****

→ Password hashing → JWT → Authentication → Authorization → Roles → CORS →
Environment variables

****16. Django****

→ Django fundamentals → Models → Views → URLs → Templates → ORM → Admin → Forms
→ Django REST Framework

****17. Testing****

→ Unit testing → pytest → API testing → Test cases → Mocking

****18. Linux****

→ Terminal → Files → Permissions → Processes → Environment variables → Basic shell
commands

****19. Docker & Deployment****

→ Docker → Images → Containers → Dockerfile → Docker Compose → Deployment → Basic
AWS

****20. Real Projects****

→ Beginner projects → CRUD application → REST API → Authentication project →
Full-stack/backend project

****21. Job Preparation****

→ Python interview questions → SQL questions → DSA → Backend questions → Project
explanation → GitHub portfolio → Resume → Mock interviews

Final Stack

****Python → SQL → Git/GitHub → Linux → REST APIs → FastAPI → PostgreSQL →
SQLAlchemy → Django/DRF → Testing → Docker → AWS → Projects → Interviews → Job****